ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Thermosphere	Ionosphere
2	**(a)** Both A and R are true and R is the correct explanation of A.	Moisture-laden
3	**(c)** land breeze	Land to sea
4	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but different layers
5	**(d)** A is false but R is true.	A false, R true
6	**(c)** thermosphere	Ionosphere
7	**(a)** Both A and R are true and R is the correct explanation of A.	Carbon sink
8	**(a)** Both A and R are true and R is the correct explanation of A.	Crossing the sea
9	**(a)** Both A and R are true and R is the correct explanation of A.	Almost all weather occurs in the troposphere, so the stratosphere gives smooth, cloud-free flying.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but R explains why temperature rises in the stratosphere, not why it falls in the troposphere.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorbs X-rays and short UV (1); Few molecules absorb high energy (1).	1+1
12	Rainfall may be concentrated (1); Poor drainage and silt (1).	1+1
13	Better forecasts (1); Helps plan crops and water (1).	1+1
14	Heat trapped by gases (1); Poor air quality hurts health (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Roots hold soil (1); Reduce flow speed (1); Trap silt (1).	1+1+1
16	Stratosphere has no weather (1); Forecasts need troposphere data (1); Cloud and rain occur there (1).	1+1+1
17	Rain varies by place (1); Local drainage and rivers (1); Better decisions (1).	1+1+1
	OR: Sowing is delayed (1); Crop yield falls (1); Food prices and rural incomes affected (1).	1+1+1
18	IMD: winter, summer, monsoon, post-monsoon (1); Rtus are six, two months each (1); Both reflect the annual cycle (1).	1+1+1
	OR: Cool sea air at day (1); Fishermen use land breeze at night (1); Comfortable climate for tourists (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Troposphere (1). (ii) Stratosphere (1). (iii) Mesosphere (1). (iv) Thermosphere (ionosphere) (1); because the lower stratosphere is calm and cloud-free, with little weather turbulence (1).	1+1+1+2
20	Greenhouse gases rise (1); Warming, floods, droughts (1); Impact on health and farming (1); Reduce carbon footprint (1); Collective policies (1).	1+1+1+1+1
	OR: Different time scales (1); Weather affects daily activity (1); Climate affects crops and housing (1); Monsoon as example (1); Forecast and planning (1).	1+1+1+1+1

Mistakes to avoid:

- **Weather** is the day-to-day state of the atmosphere; **climate** is the average weather over 30 years or more. A single rainy day is not climate.

- Sea breeze blows from sea to land in the **day** (land is warmer, low pressure on land); land breeze blows from land to sea at **night** and is weaker. Do not reverse them.
- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.
- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.
- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** weather of the place	Short-term state of the atmosphere
2	**(a)** north-west India (Rajasthan)	Retreat
3	**(b)** Kerala	Starts in the south
4	**(a)** global warming	Heat trapped
5	**(a)** Both A and R are true and R is the correct explanation of A.	Human factors
6	**(a)** Both A and R are true and R is the correct explanation of A.	Differential cooling
7	**(d)** Driving alone in a petrol car	Fossil fuels
8	**(c)** Frequent use of petrol car	Fossil fuels
9	**(c)** A is true but R is false.	Land heats faster than the sea, creating low pressure over land; so R is false.
10	**(d)** A is false but R is true.	Trapping heat warms the Earth, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Predict rain timing (1); Plan sowing and harvest (1).	1+1
12	Sea moderates temperature (1); Breezes cool by day and warm by night (1).	1+1
13	Shows where emissions arise (1); Helps plan reduction (1).	1+1
14	Weak old embankments (1); Houses built close to rivers (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat stress and disease (1); Floods and drought damage crops (1); Livelihoods suffer (1).	1+1+1
16	Rain pattern follows seasons (1); Planting is timed by season (1); Tradition and science combine (1).	1+1+1
17	Forecasts have limits (1); People must act on warnings (1); NDMA guidelines help (1).	1+1+1
	OR: Air thinner at height (1); Less weight above (1); Pressure falls rapidly with altitude (1).	1+1+1
18	Gases trap heat (1); Aerosols reflect sunlight (1); Change temperature and rain patterns (1).	1+1+1
	OR: Heavy rain upstream (1); Weak embankments and settlements near river (1); Low-lying lands flooded (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Season (1). (ii) June to September (1). (iii) The Tamil Nadu coast (1). (iv) The wind direction reverses with the season, blowing from sea to land in summer and from land to sea in winter (1); a weak monsoon can cause drought and affect farming (1).	1+1+1+2
20	Insolation drives temperature (1); Warm air rises, low pressure (1); Winds carry moisture (1); Precipitation follows (1); Averages over 30 years form climate (1).	1+1+1+1+1
	OR: Krishiparashara and Brihatsamhita (1); Nakshatra observation (1); Kalidasa's monsoon account (1); NMM and Mission Mausam (1); Joint use helps farmers (1).	1+1+1+1+1

Mistakes to avoid:

- **Weather** is the day-to-day state of the atmosphere; **climate** is the average weather over 30 years or more. A single rainy day is not climate.
- Sea breeze blows from sea to land in the **day** (land is warmer, low pressure on land); land breeze blows from land to sea at **night** and is weaker. Do not reverse them.
- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.

- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.
- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** A is false but R is true.	A false: different time scale
2	**(c)** Troposphere, stratosphere, mesosphere, thermosphere, exosphere	Order from the surface
3	**(c)** climate	Long-term pattern
4	**(b)** Solar panels	Renewable energy
5	**(a)** Both A and R are true and R is the correct explanation of A.	Weather-free
6	**(d)** divide the Moon's path into 27 parts	Traditional astronomy
7	**(b)** Insolation decreases	Sun angle
8	**(b)** internal pressure equals external	Balanced
9	**(a)** Both A and R are true and R is the correct explanation of A.	Moisture gathered over the Bay of Bengal is dropped on the coast of Tamil Nadu.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Nitrogen (about 78 per cent) is the most abundant; the share of oxygen is a true fact but does not explain it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorb heat slowly (1); Water volume rises with melting ice (1).	1+1
12	Weather varies daily (1); Climate needs 30-year data (1).	1+1
13	Preparedness for climate change (1); Using better forecasts (1).	1+1
14	Traps heat (1); Shields from radiation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Clean water and sanitation (1); Drain standing water (1); Medical camps (1).	1+1+1
16	Fossil fuels add CO2 (1); Forests absorb CO2 (1); Net balance decides footprint (1).	1+1+1
17	Warm humid air feels sticky (1); Sweat does not evaporate (1); Coastal cities in summer (1).	1+1+1
	OR: Over land it is dry (1); Over Bay of Bengal it picks up moisture (1); Then rains on east coast (1).	1+1+1
18	Save electricity and water (1); Encourage walking, cycling (1); Reduce plastic and plant trees (1).	1+1+1
	OR: Strong heating at equator lifts air (1); Poles are colder (1); Higher tropopause at equator (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Any two of Satluj, Beas, Ravi (1). (ii) Monsoon (1). (iii) Carbon dioxide (or methane) (1). (iv) Weak embankments or building near rivers (1); planting trees, cycling or using solar energy (1).	1+1+1+2
20	Troposphere for air and weather (1); Stratosphere's ozone filters UV (1); Mesosphere burns meteors (1); Thermosphere reflects radio waves (1); Gravity holds the whole blanket (1).	1+1+1+1+1
	OR: Personal steps reduce footprint (1); Governments set policy (1); Industries must cut emissions (1); Collective action is needed (1); Awareness spreads change (1).	1+1+1+1+1

Mistakes to avoid:

- **Weather** is the day-to-day state of the atmosphere; **climate** is the average weather over 30 years or more. A single rainy day is not climate.
- Sea breeze blows from sea to land in the **day** (land is warmer, low pressure on land); land breeze blows from land to sea at **night** and is weaker. Do not reverse them.
- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.

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